Computer and Data Communications

Chapter 1_1

Introduction

โดย เศรษฐา ตั้งค้าวานิช

Objectives

- Define the basic terminology of computer networks.
- Recognize the individual components of the big picture.
- Outline the basic network configuration.

Introduction

- Who today has *not* used a computer network?
- Mass transit, interstate highways, 24-hour bankers, grocery stores, cable television, cellular telephones, most businesses and schools, and other retail outlets can support some form of computer networks.




Road Traffic Information


ATM




PRODUCE
99¢
249









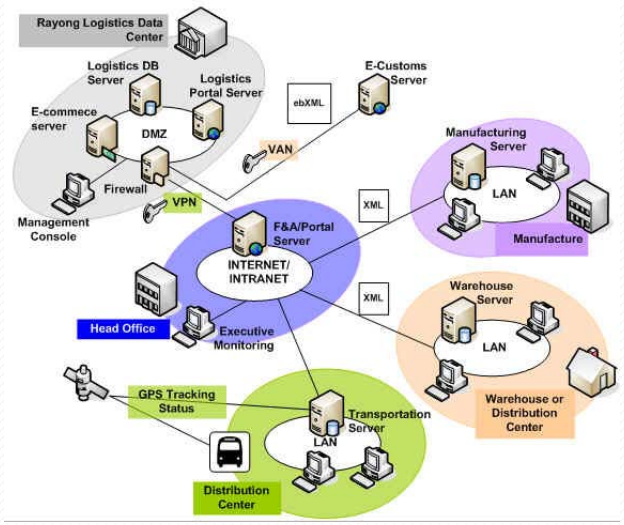




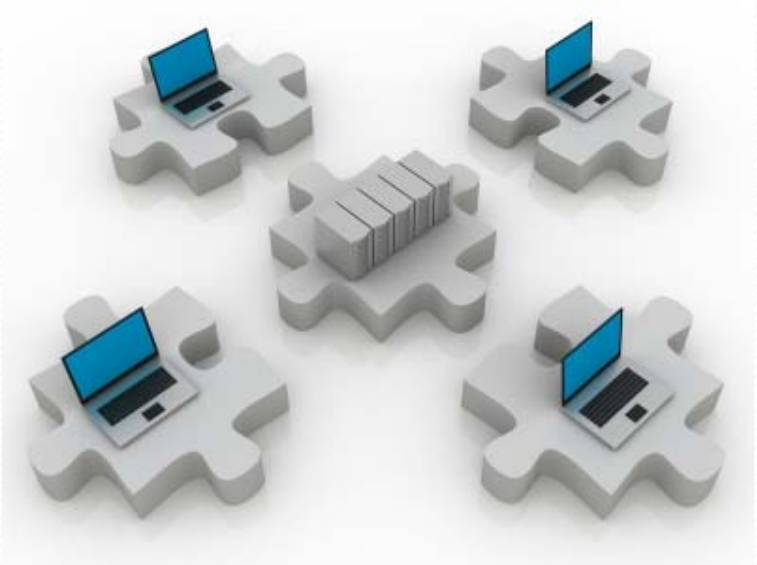


The Language of Computer Networks









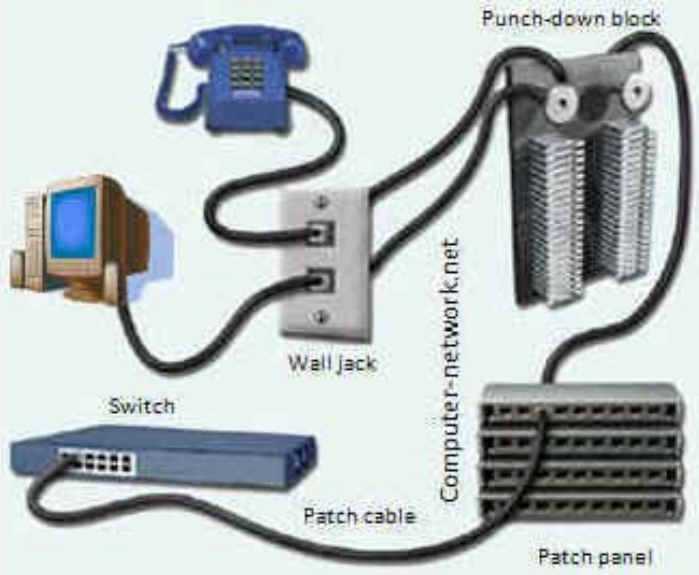
The Language of Computer Networks

- **Computer network** is an interconnection of computers



The Language of Computer Networks

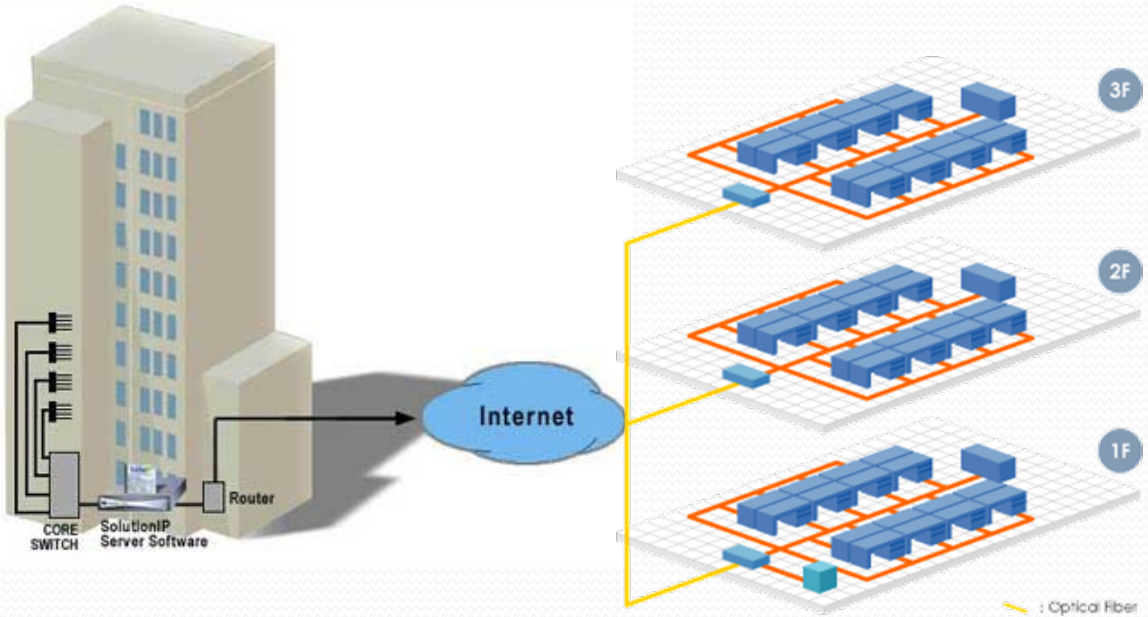
- Computer networks use either wire or radiowave (termed wireless)





The Language of Computer Networks

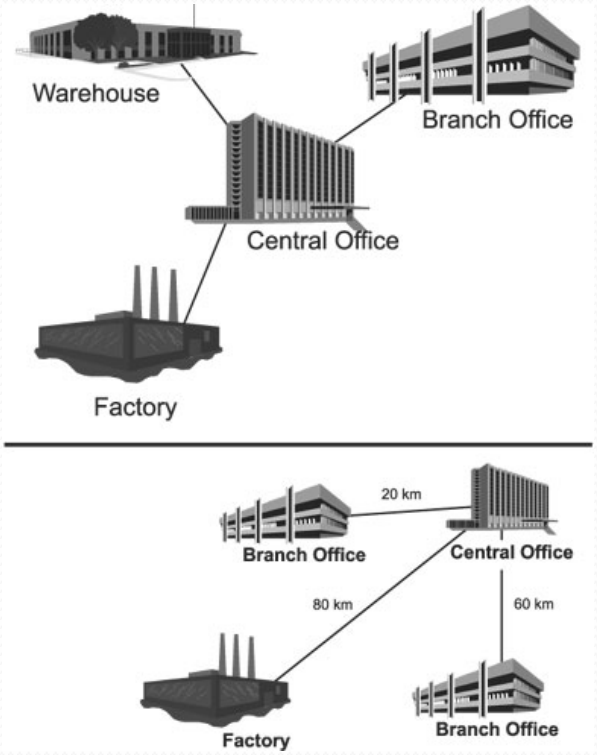
- Network is categorized by geographic size .
- Networks that are small in geographic size are called local area nerworks (**LANs**)
- LANs cover a room, a building, a campus.

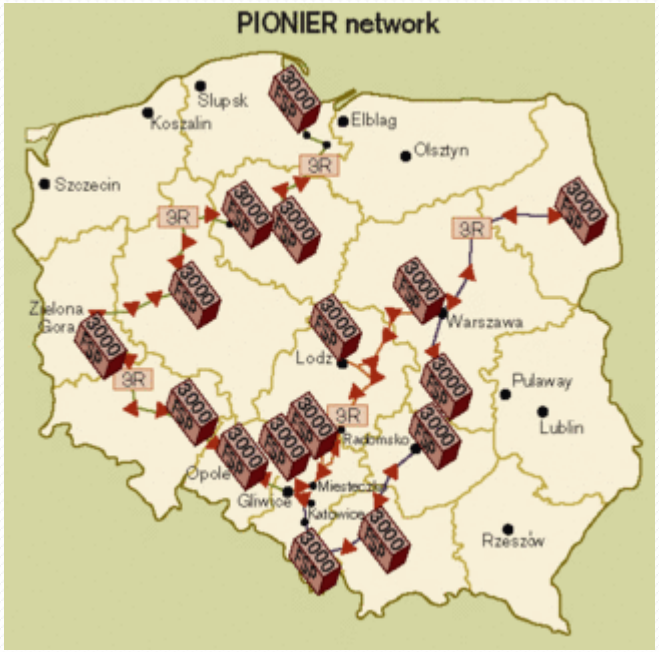




The Language of Computer Networks (cont)

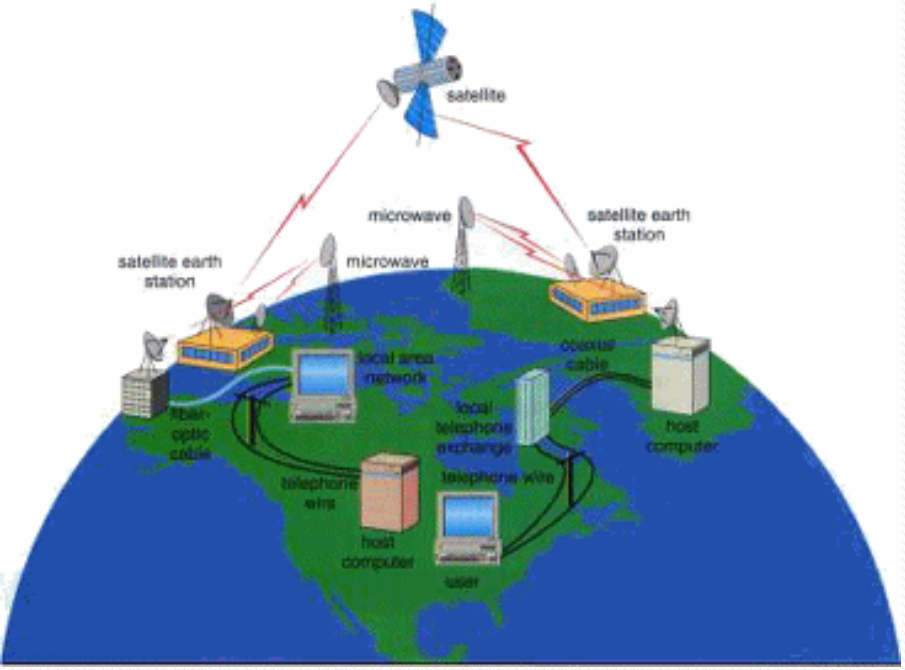
- Metropolitan area networks (**MANs**) cover an area up to 30 miles- approximately the area of the typical city.





The Language of Computer Networks (cont)

- Wide area networks (**WANs**) cover an area of multiple countries or the world.





The Language of Computer Networks (cont)

- Personal area networks(**PANs**) cover an area of several meters between wireless devices (including laptop, Personal digital assistants PDAs.)





The Language of Computer Networks (cont)

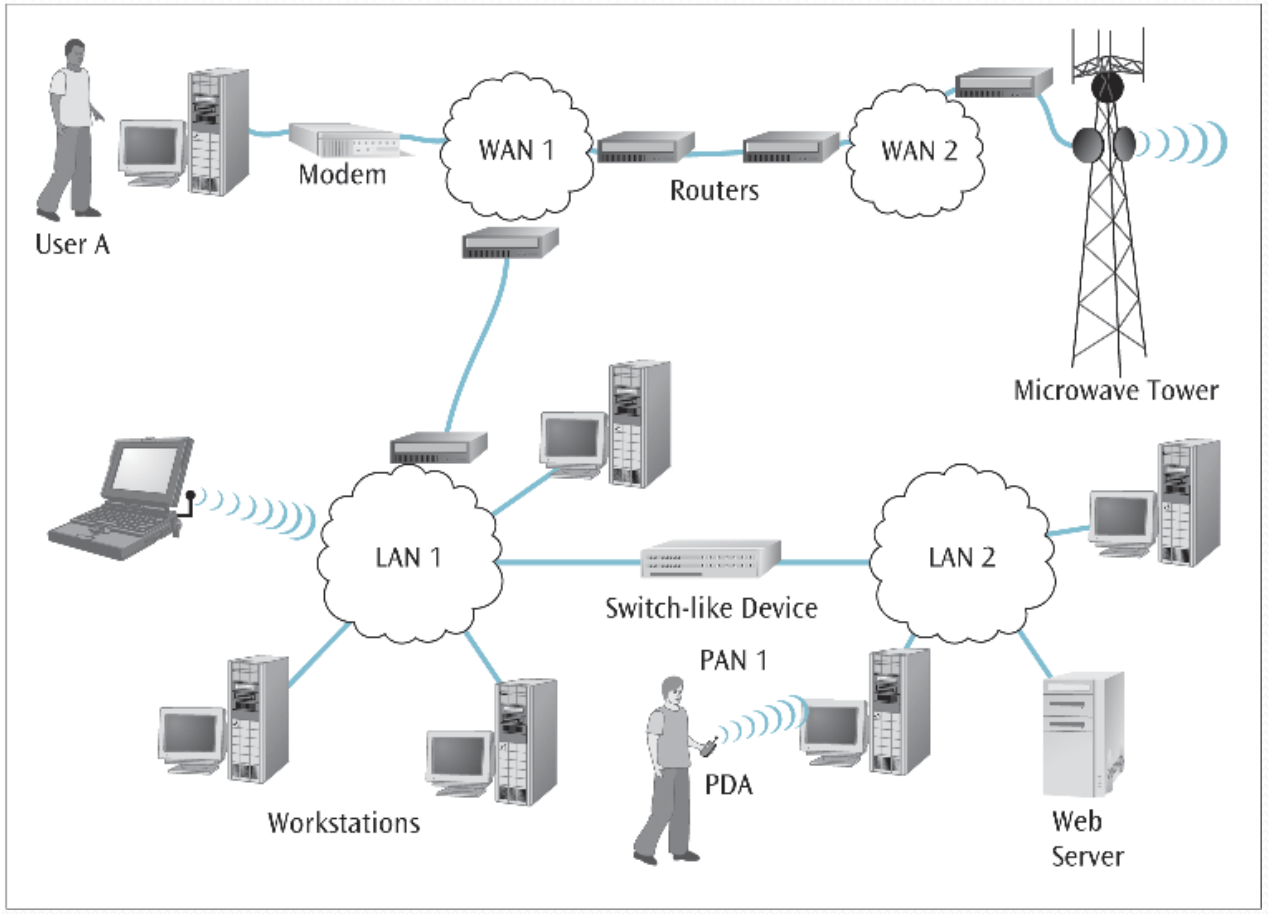
- **Data communication** is the transfer of digital or analog data using digital or analog signal.
- **Telecommunication** is the study of telephones and systems
- **Voice network** - a network that transmits telephone signals
- **Data network** - a network that transmits computer data
- **Network management** - the design, installation, and support of a network and its hardware and software

The Language of Computer Networks (cont)

- **Modems** is devices modulating digital data to analog signal.
- **Codec** or **digitizer** is devices converting analog data to digital signal.
- **Multiplexing** is transmitting multiple signals on one media.

The Big Picture of Networks

Figure 1-1
An overall view of the interconnection between different types of networks



The Big Picture of Networks

- **Workstations** are personal computers or microcomputers, where users reside.





The Big Picture of Networks

- **Servers** are the computer that store the network software and shared or private user files.





The Big Picture of Networks

- **Bridges** are the devices connecting between LANs

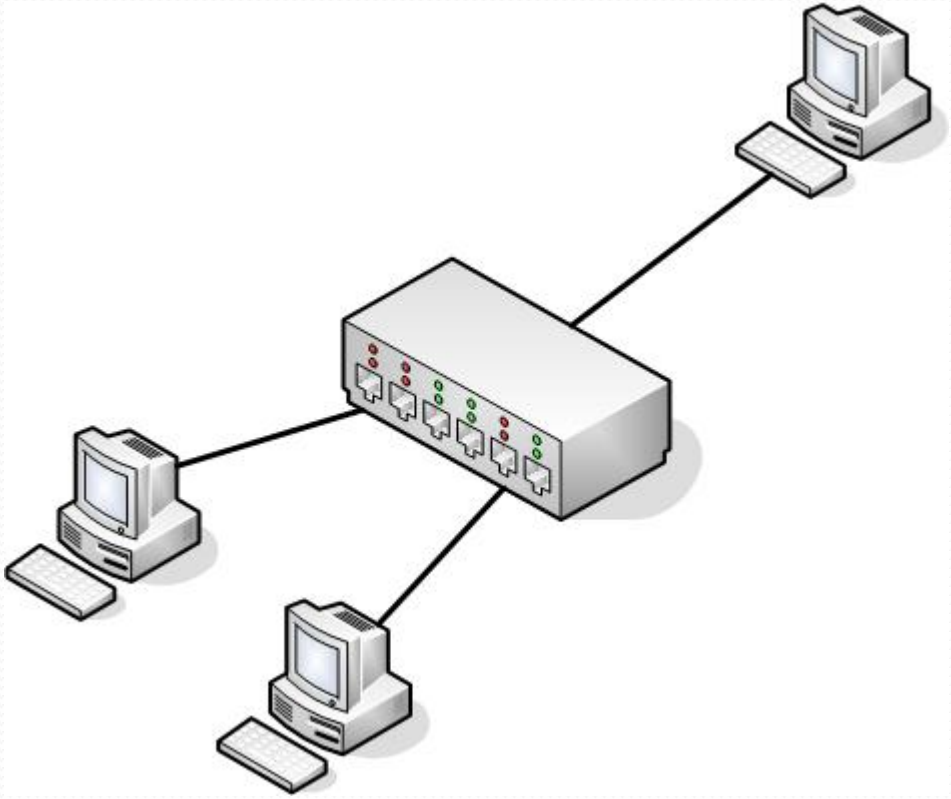




The Big Picture of Networks

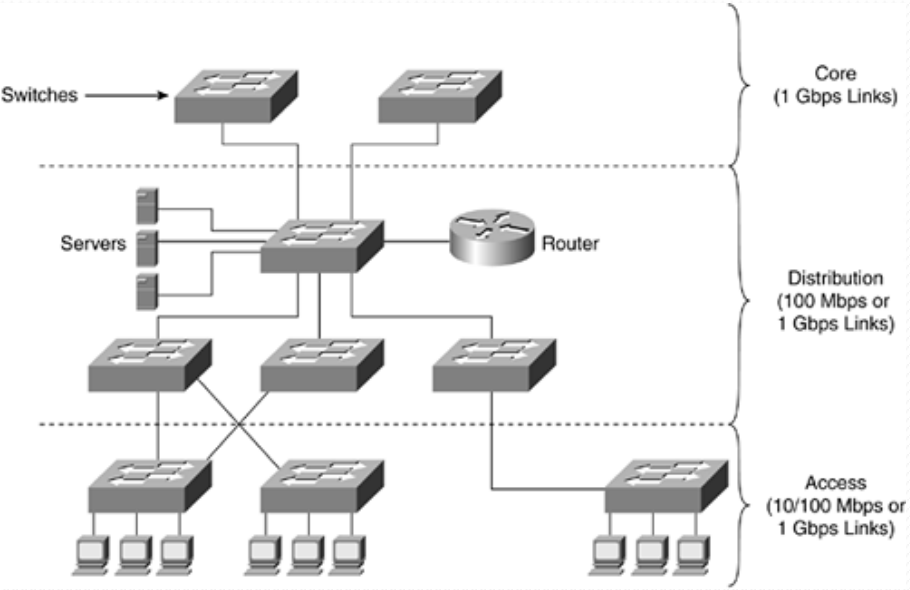
- **Hubs** are the devices connecting between Workstations.



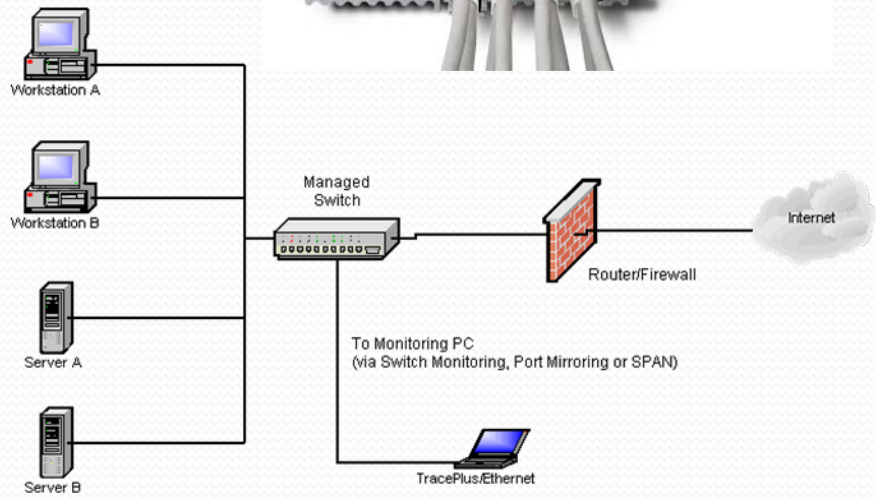


The Big Picture of Networks

- **Switches** are more advanced devices replacing hubs and bridge.



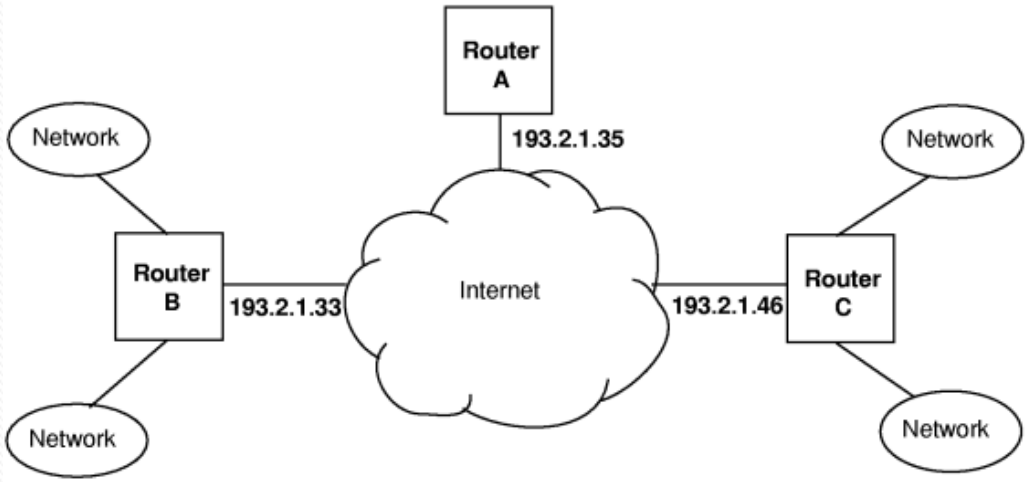




The Big Picture of Networks

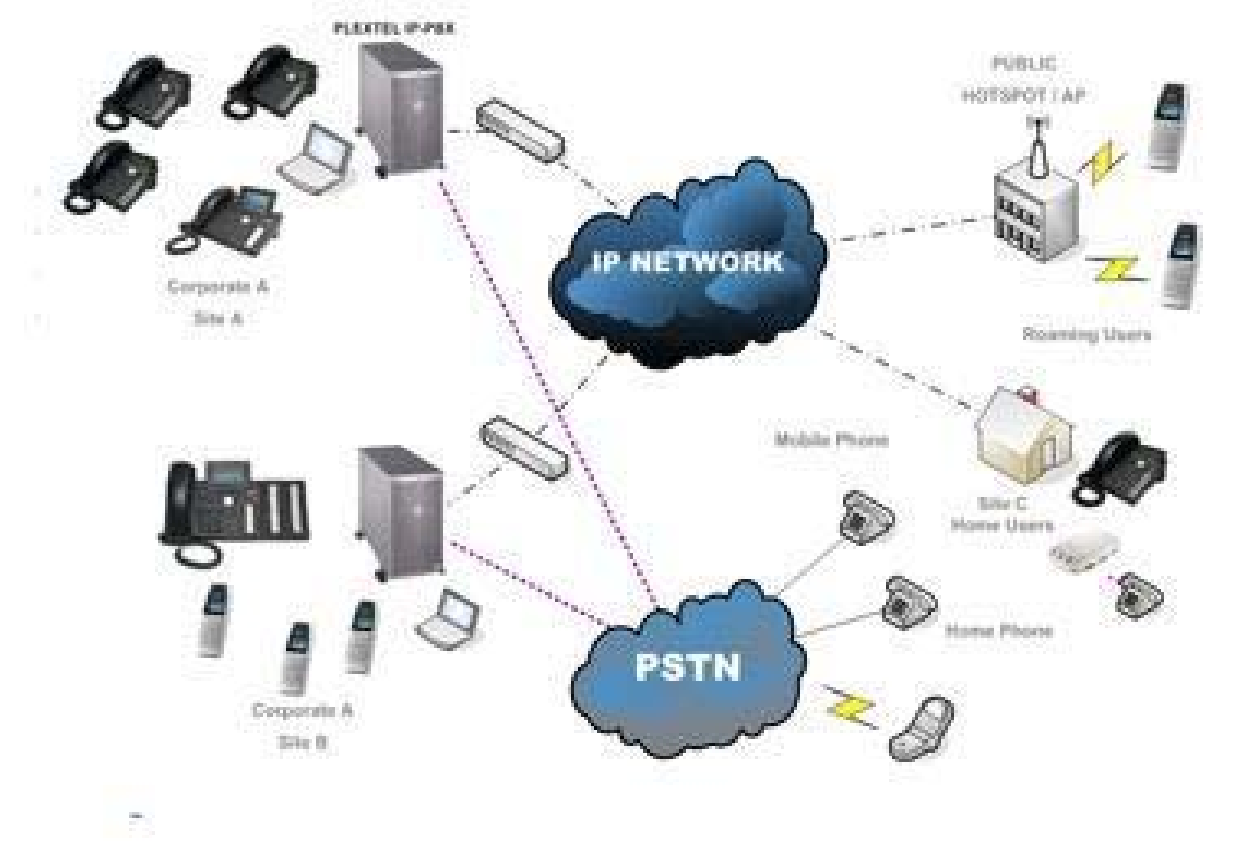
- **Routers** are the devices connecting between LANs and WANs. (LAN-WAN and WAN-WAN)





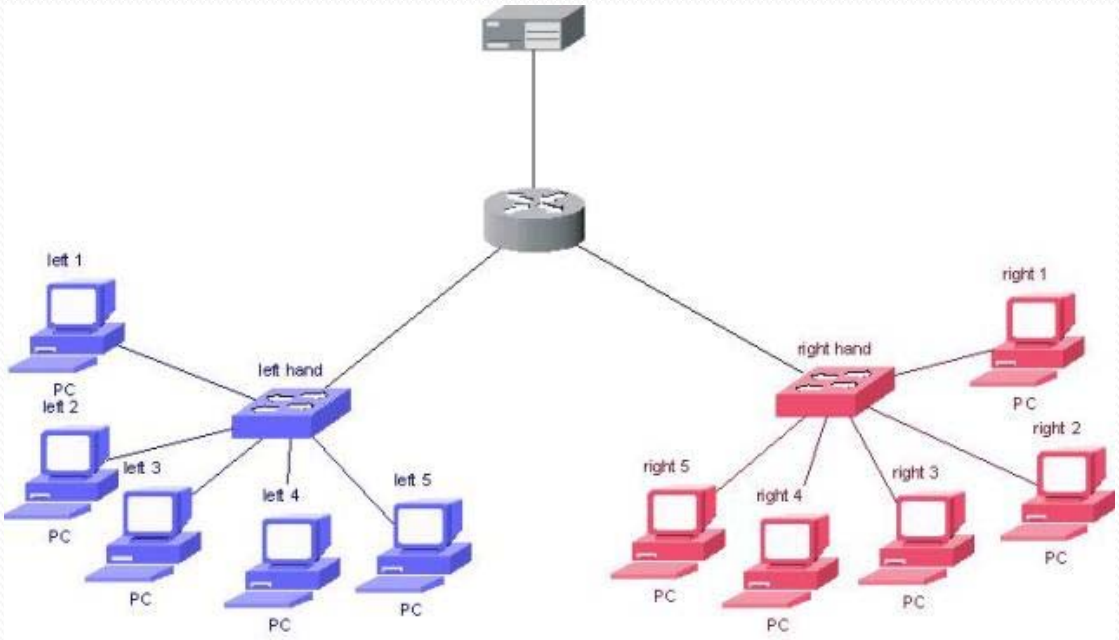
The Big Picture of Networks

- **Nodes** are computing devices that allow workstation connect to the networks



The Big Picture of Networks

- **Subnet** is the collection of nodes and transmission line



Basic Configuration

- Terminal-to-mainframe computer
- Microcomputer-to-mainframe computer
- Microcomputer-to-local area network

  - Client /server system
- Microcomputer-to-Internet
- Local area network-to-local area network
- Personal area network-to-workstation

Basic Configuration

- Local area network-to-metropolitan area network
- Local area network-to-wide area network
- Sensor-to local area network
- Satellite and microwave
- Wireless telephone

Terminal/Microcomputer to Mainframe Computer

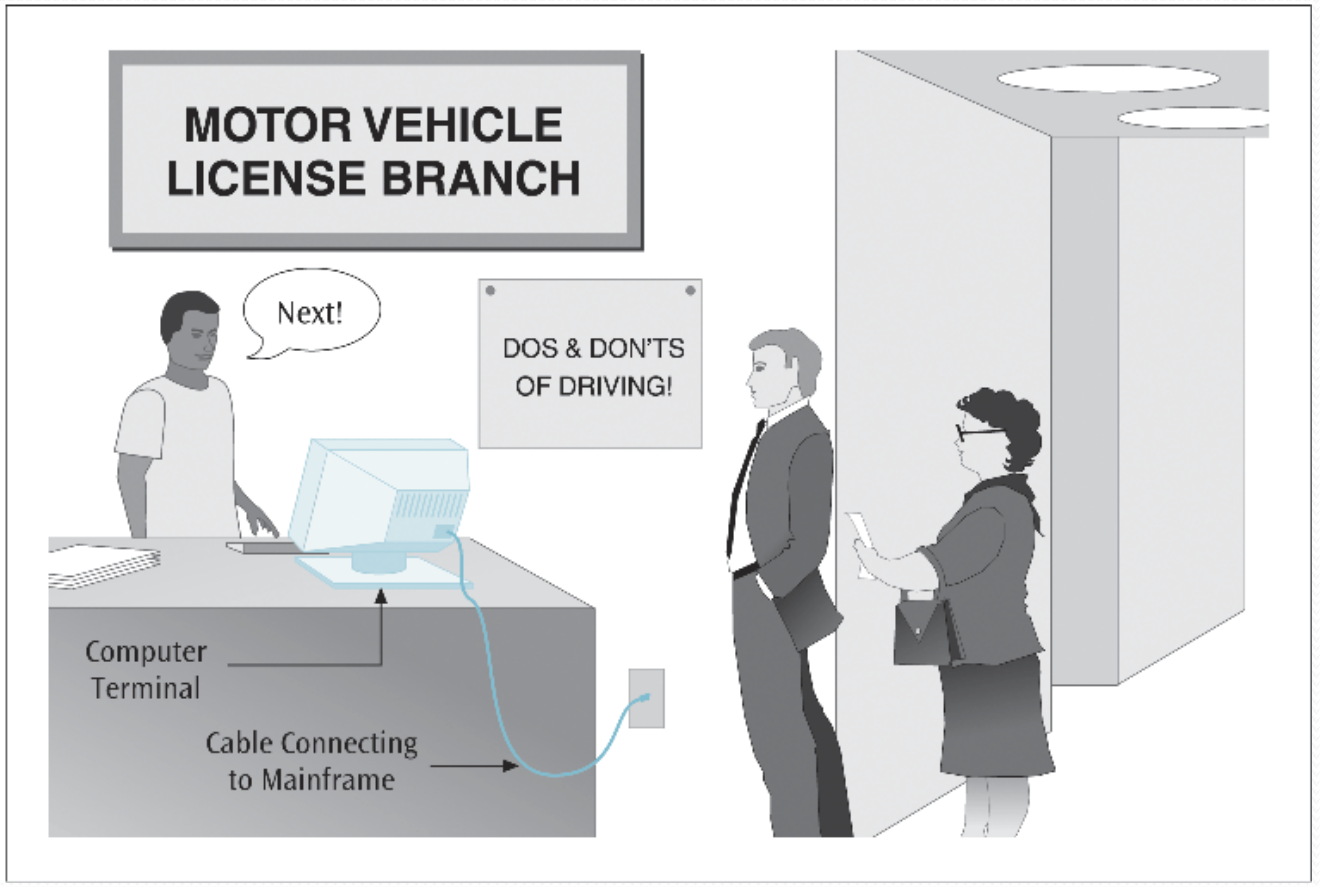
- Predominant form in 60s and 70s
- Still used in many types of businesses for data entry and data retrieval.
- Usually involves a low-speed connection.

Terminal/Microcomputer to Mainframe Computer

- Terminal is only used for entering and displaying data with no long-term storage.
- Microcomputer can download information from mainframe, perform the data and then upload data to mainframe.

Terminal/Microcomputer to Mainframe Computer

Figure 1-2
Using a terminal to perform a text-based input transaction



Microcomputer to Local Area Network

- Highly common throughout business and academic environments, and now even homes
- Typically a medium- to high-speed connection
- Microcomputer requires a NIC (network interface card)
- NIC connects to a hub-like device

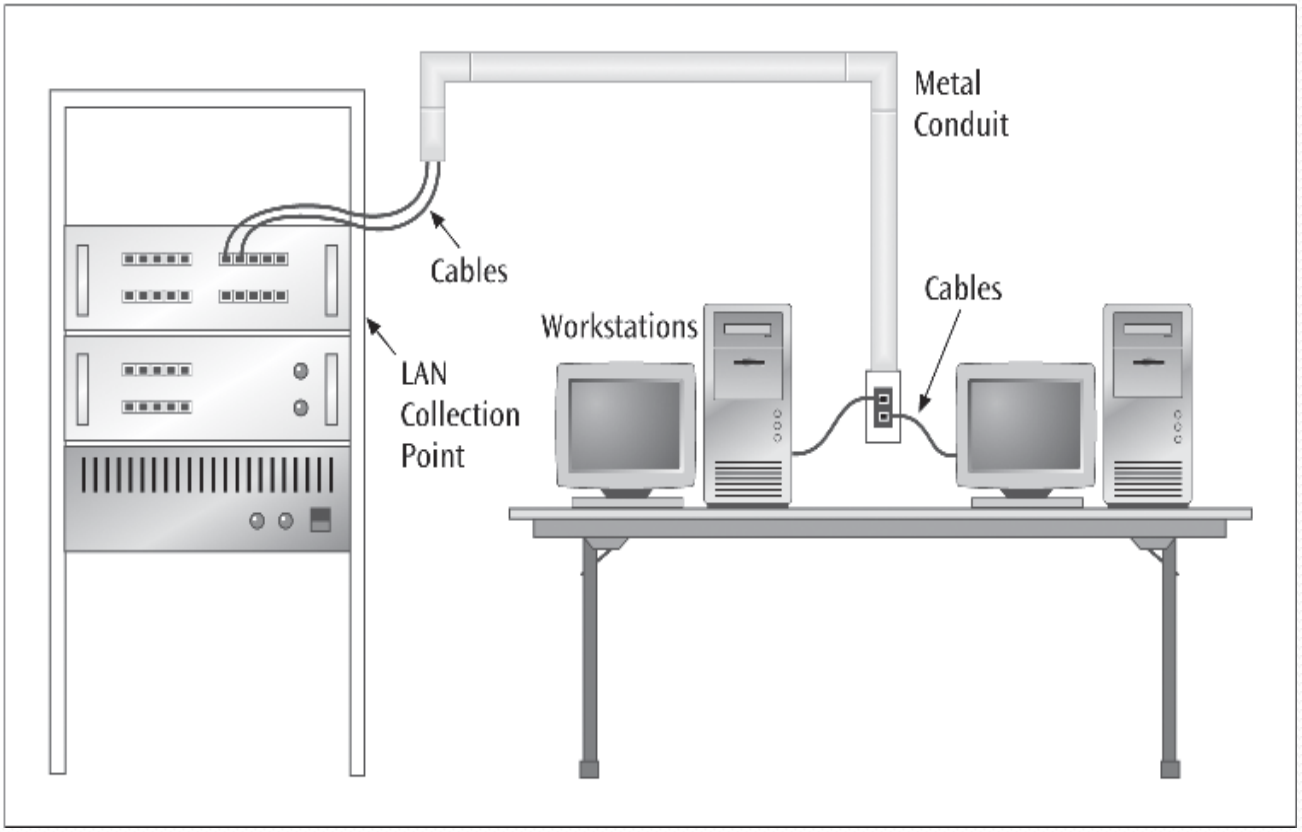
Microcomputer to Local Area Network

- Microcomputer to Local Area Network configuration is the client/server system.
- Users at microcomputer called client machine issue the request for data or service to server.
- The server containing a large repository of data and programs fill the request and return the result to the clients

Microcomputer to Local Area Network

Figure 1-3

A microcomputer lab, showing the cabling that exits from the back of a workstation and runs to a LAN collection point

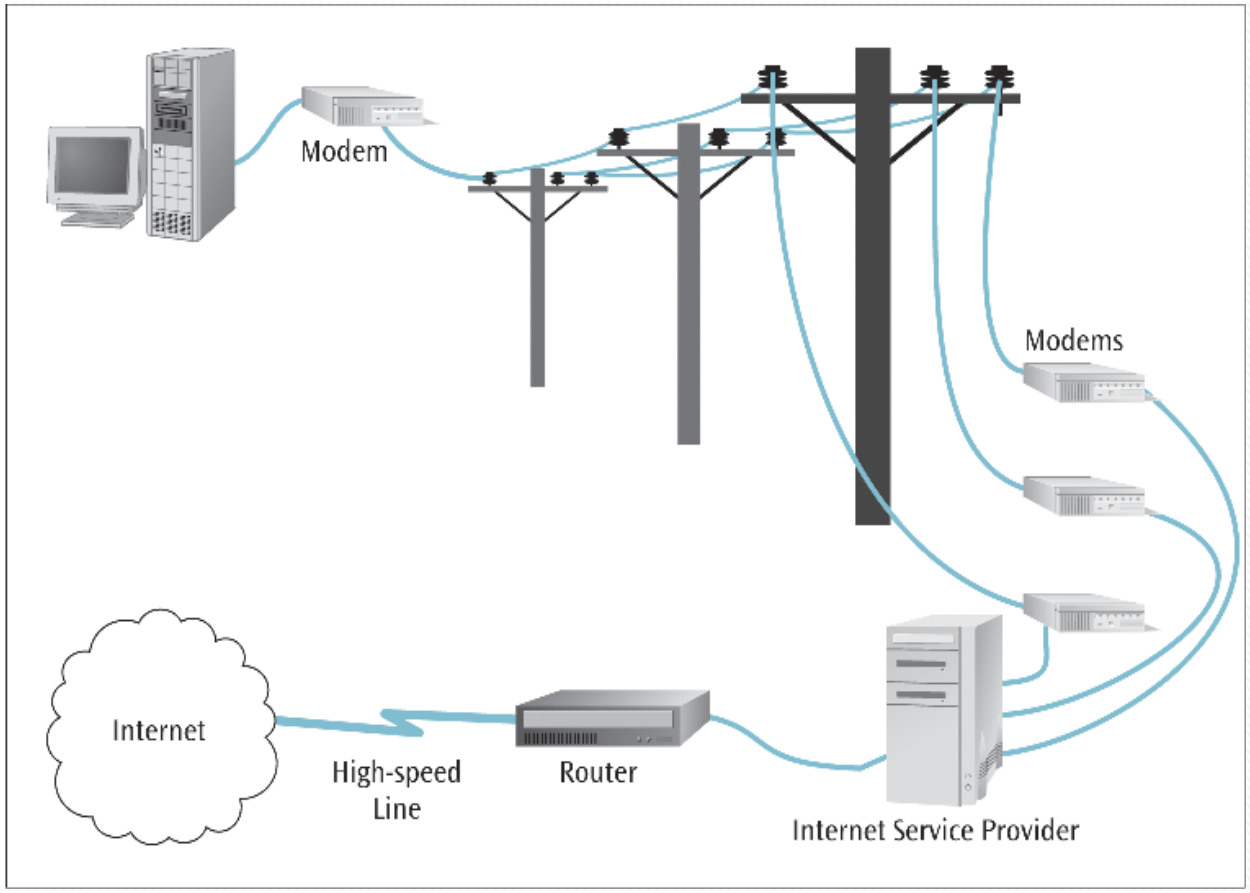


Microcomputer to Internet

- Very popular with home users and some small businesses
- Typically a dial-up modem is used to connect user's microcomputer to an Internet service provider.
- Newer technologies such as DSL and cable modems are replacing modems.

Microcomputer to Internet

Figure 1-4
A microcomputer/ workstation sending data over a telephone line to an Internet service provider and onto the Internet

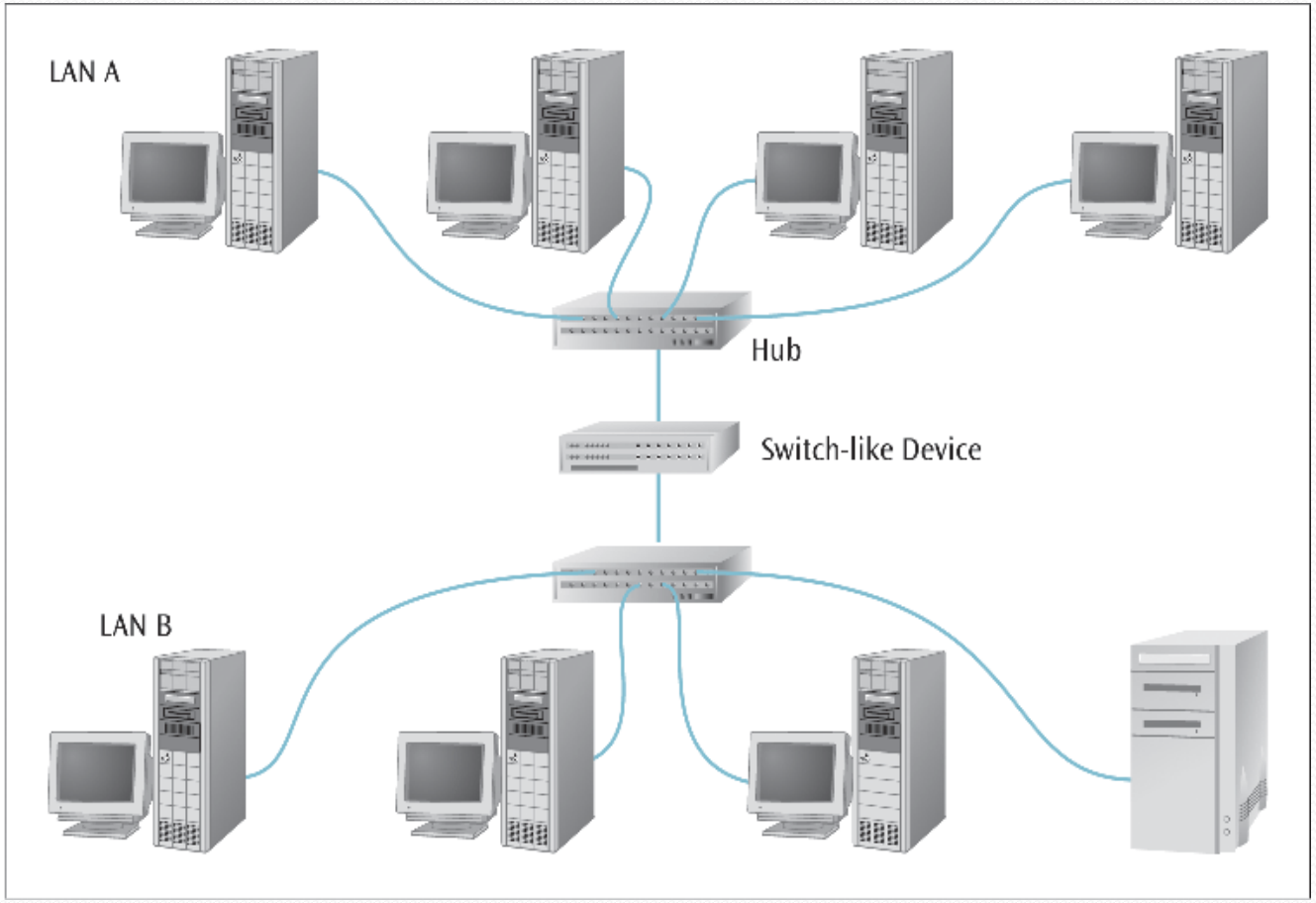


Local Area Network to Local Area Network

- Found in businesses and schools that have two or more LANs and a need for them to intercommunicate.
- A bridge-like device (such as a switch) is typically used to interconnect LANs.
- Bridge-like device can *filter* frames.

Local Area Network to Local Area Network

Figure 1-5
Two local area networks connected by a switch-like device

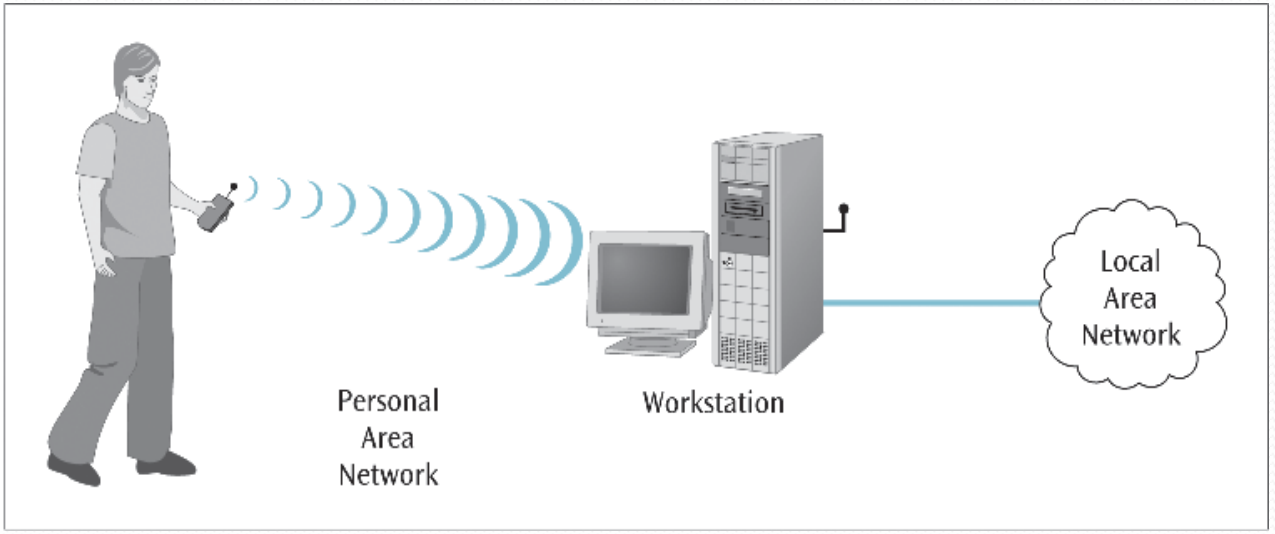


Personal Area Network to Workstation

- Interconnects wireless devices such as PDAs, laptops, and music playback devices
- Used over a short distance such as a few meters

Personal Area Network to Workstation

Figure 1-6
A user transferring data from a personal digital assistant to a workstation attached to a local area network via a personal area network

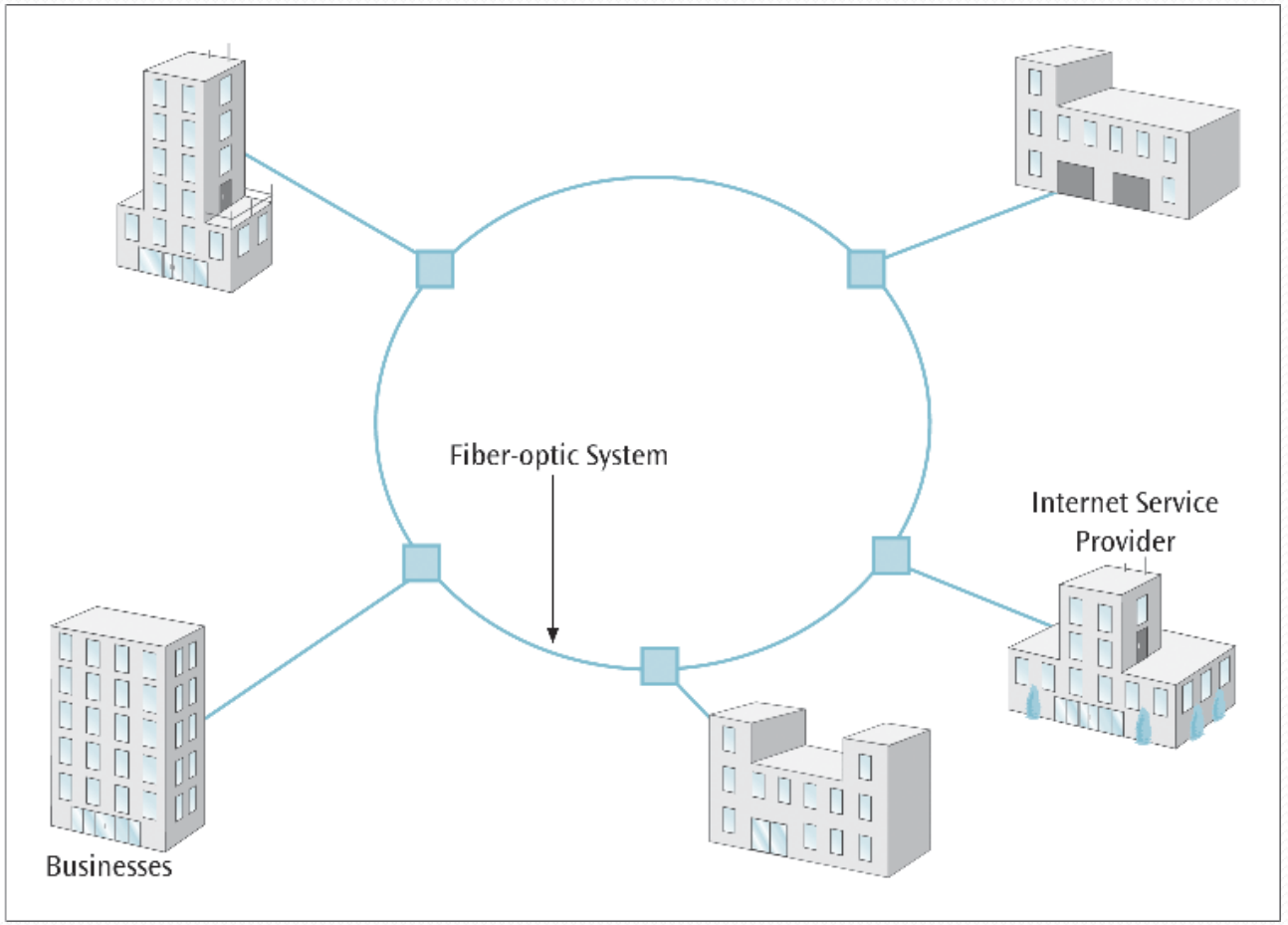


Local Area Network to Metropolitan Area Network

- Used to interconnect companies (usually their local area networks) to networks that encompass a metropolitan city.
- High speed networks with redundant circuits are the networks that interconnect multiple sites within a close geographic region.

Local Area Network to Metropolitan Area Network

Figure 1-7
Businesses interconnected within a large metropolitan area via a metropolitan area network

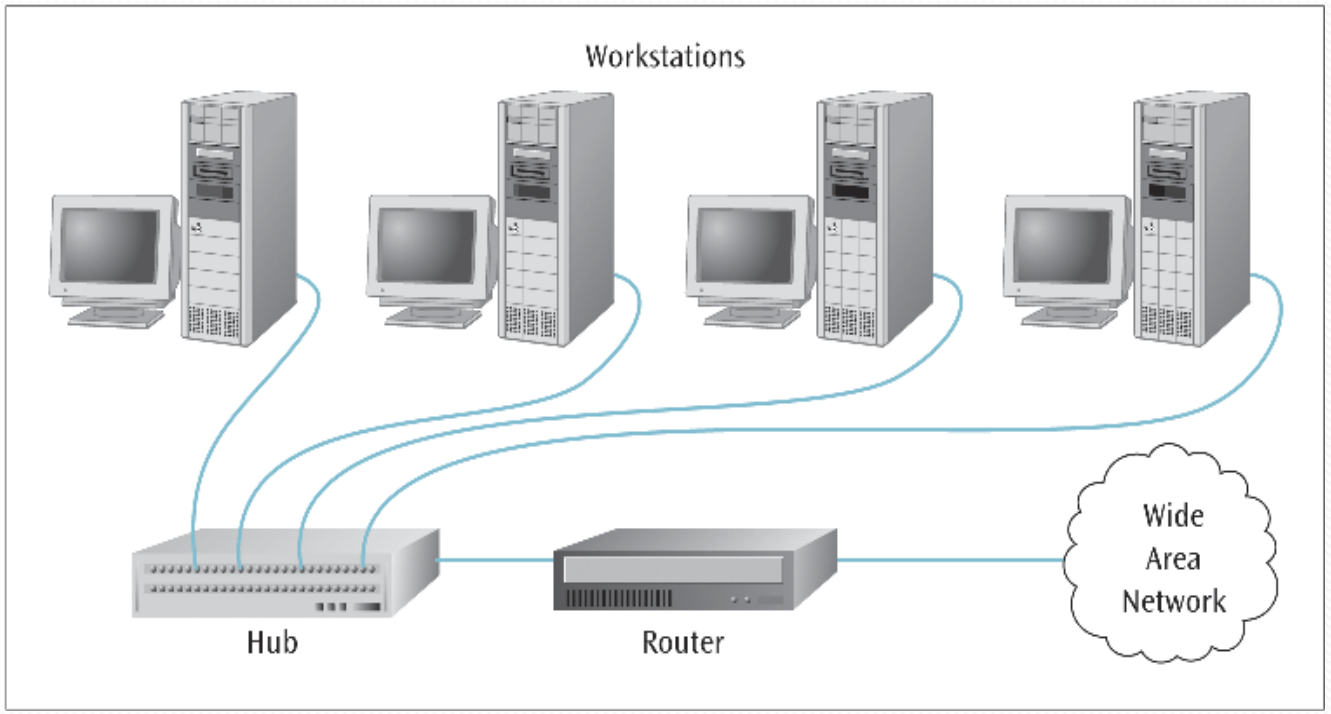


Local Area Network to Wide Area Network

- One of the most common ways to interconnect a user on a LAN workstation to the Internet (a wide area network).
- A router is the typical device that performs LAN to WAN connections.
- Routers are more complex devices than bridges/switches because they convert data from LANs into the data bound for WANs

Local Area Network to Wide Area Network

Figure 1-8
Local area network-to-wide area network configuration

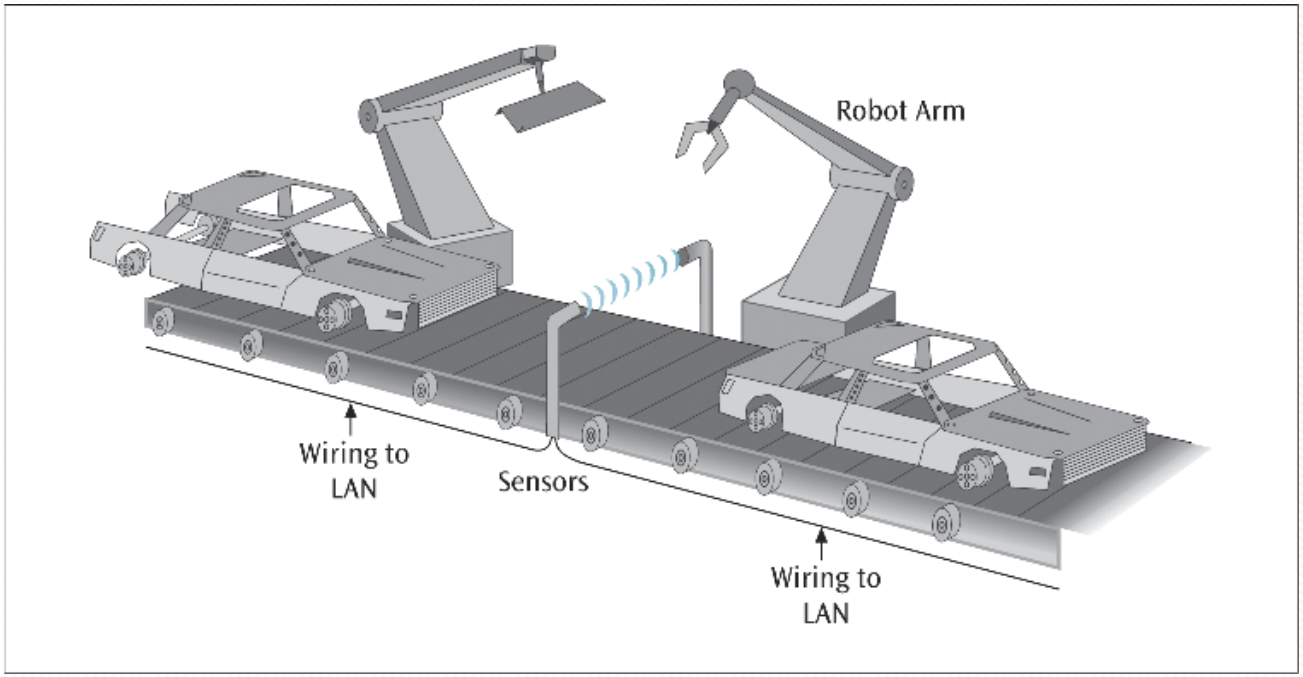


Sensor to Local Area Network

- Not all local area networks deal with microcomputer workstations.
- Often found in industrial and laboratory environments.
- Assembly lines and robotic controls depend heavily on sensor-based local area networks.

Sensor to Local Area Network

Figure 1-9
An automobile moves down an assembly line and triggers a sensor

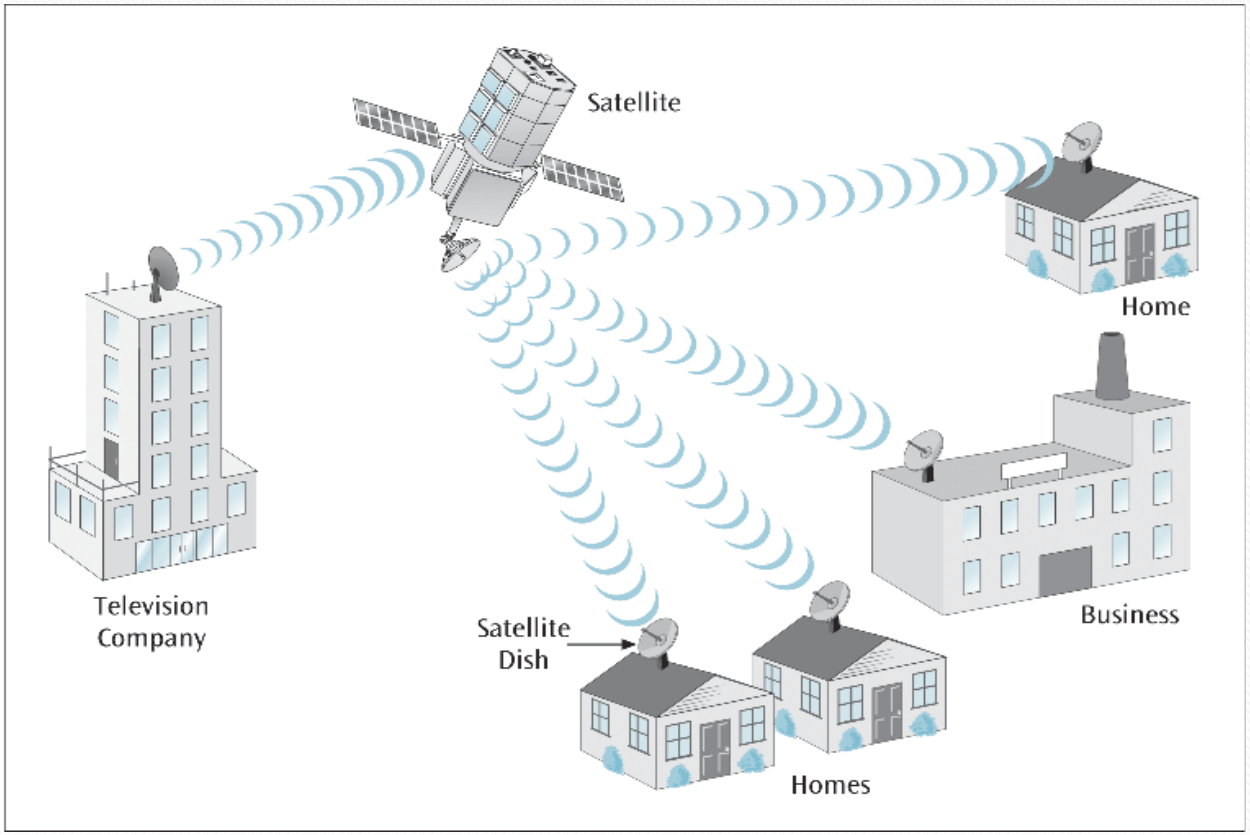


Satellite and Microwave

- Long distance wireless connections
- Many types of applications including long distance telephone, television, radio, long-haul data transfers, and wireless data services.

Satellite and Microwave

Figure 1-10
Example of a television company using a satellite system to broadcast television services into homes and businesses

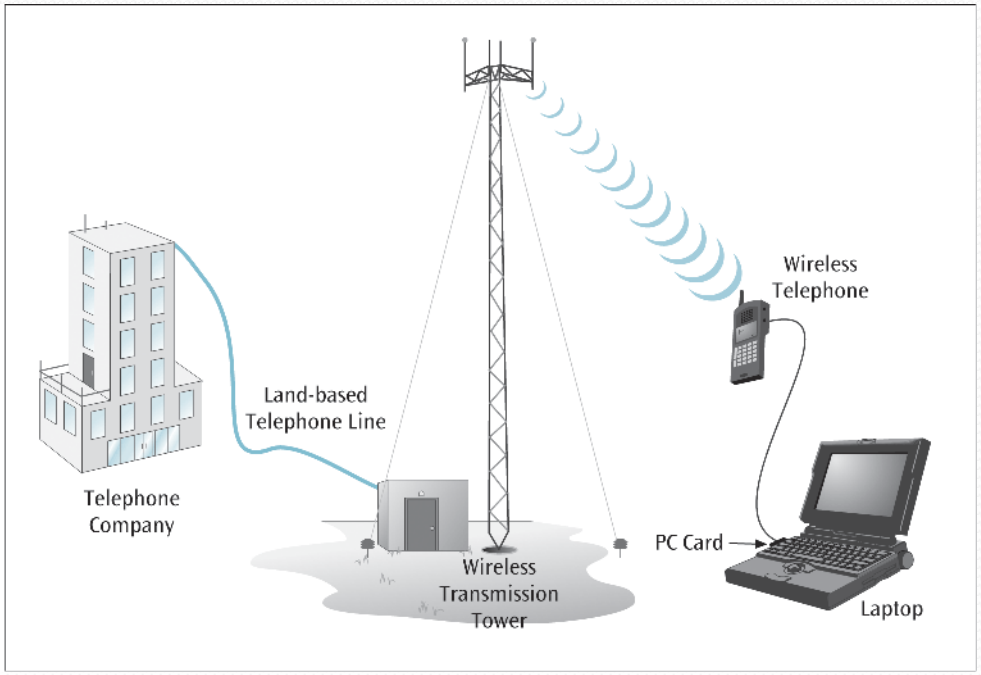


Wireless or Cellular Telephones

- Constantly expanding market over the world
- Second generation PCS(Personal Communication System) services available in most areas and under many types of plans
- Next generation services beginning to replace PCS phones

Wireless or Cellular Telephones

Figure 1-11
An example of a laptop computer connected to a wireless telephone system to transmit and receive data



Telephone to Network

- Telephone systems are ubiquitous and can now carry more data than voice
- Common configuration – telephone connected to POTS(Plain Old Telephone System)
- Newer configuration – telephone to LAN via gateway (VoIP)

Review Questions

- จงเขียนคำจำกัดความของ

- Computer network
- Data communication
- Telecommunication
- Local area network
- Metropolitan area network
- Personal area network
- Wide area network

Review Questions

- อะไรคือความสัมพันธ์ระหว่างของ Subnet และ nodes

- จงตัวอย่างของ Application ของ

- Terminal-to-mainframe computer
- Microcomputer-to-mainframe computer
- Sensor-to local area network

Computer and Data Communications

Chapter 1_2

Network Architecture Model

โดย เศรษฐา ตั้งค้าวานิช

Objectives

- Cite the reasons for using a network model.
- List the layer of the OSI model and describe the duties of each layer.

Network Architecture Model

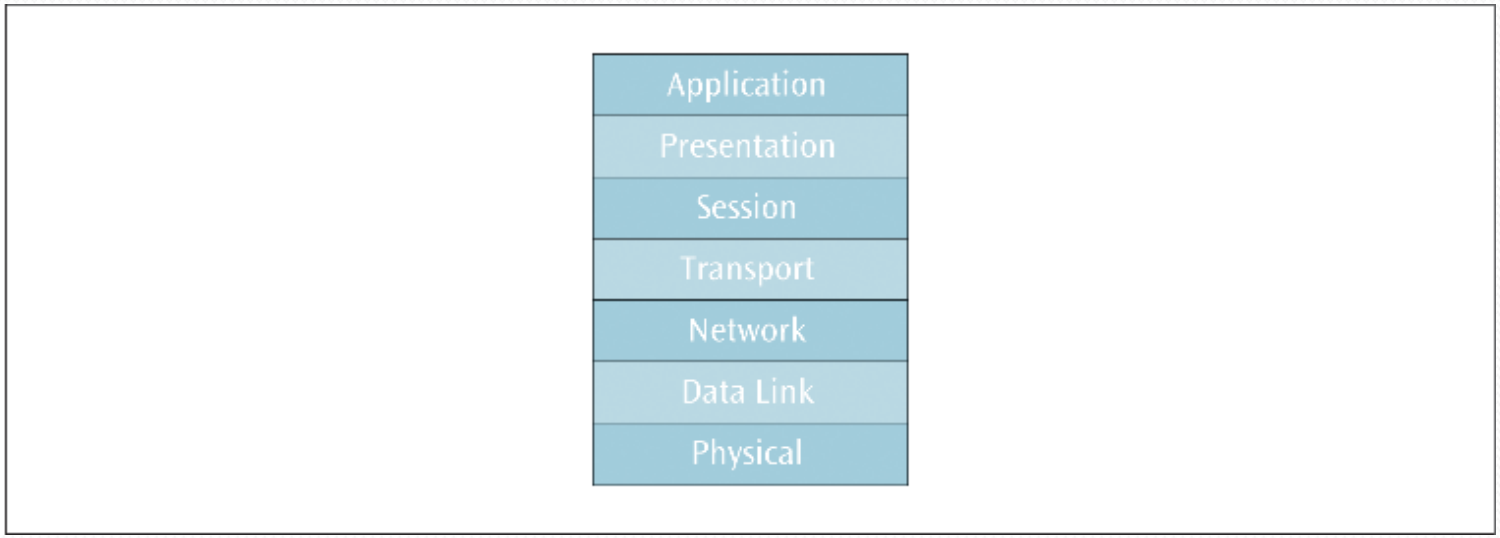
- A reference model that describes the layers of hardware and software necessary to transmit data between two points or for multiple devices / applications to interoperate
- Reference models are necessary to increase the likelihood that different components from different manufacturers will converse

Network Architecture Model

- Network architecture model or communication model is divided into the layers
- The layers define a model for functions or services that need to be perform
- Each layer in the model define what services either hardware or software or both provide.
- The two most common models are Open System Interconnection (OSI) and the Internet model (or TCP/IP)

Open System Interconnection Model (OSI)

Figure 1-12
The seven layers of the OSI model



Open System Interconnection Model (OSI)

- OSI consists of seven layers
- **The Application layer** is where the application using the network resides
- The examples of applications are electronic mail , file transfer, remote login and etc.

Open System Interconnection Model (OSI)

- **The Presentation layer** performs a series of miscellaneous functions necessary for presenting the data package properly to the sender or receiver
- **The Presentation layer** performs ASCII to non-ASCII conversion, encrypt and decrypt secure documents and compress data in small units.

Open System Interconnection Model (OSI)

- **The Session layer** is responsible for establishing sessions between users and for **token management** and **synchronization point** where are backup point in case of errors or failures.
- **The transport layer** provide end-to-end error-free network and flow control to ensure that data arrives at the final destination error free (no transmission error, the data in the same order as the transmitted data and no duplication of data).

Open System Interconnection Model (OSI)

- **The Network layer** is responsible for creating, maintaining and ending network connection.
- **The Network layer** transfers a data packet from node to node within the network.
- By network addressing, the network layer can define path of data and also control congestion.

Open System Interconnection Model (OSI)

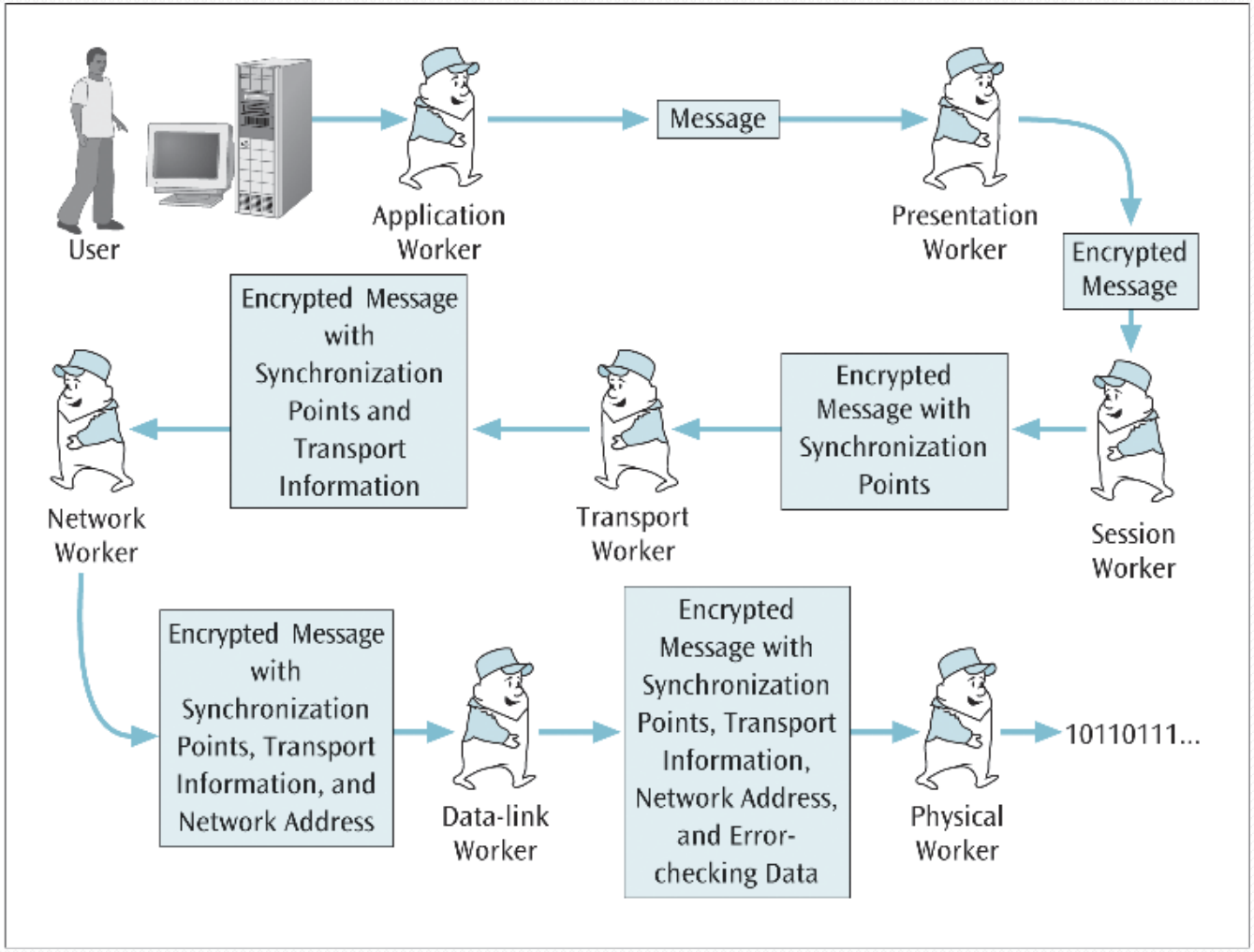
- **The data link layer** is responsible for taking the data from the network layer and transforming a cohesive unit called a **frame**
- The data link layer provide flow and error control between two nodes
- The frame contains data from network layer,the beginning and ending frame and a space for address and control information (error and flow control).

Open System Interconnection Model (OSI)

- **The physical layer** handles the transmission of bits over a communication channel.
- **The physical layer** handles voltage level, plug and connector dimensions, pin configuration and the other electrical and mechanical issues (included encoding and modulation technique).

Open System Interconnection Model (OSI)

Figure 1-13
Network workers perform their job duties at each layer in the model



Review Questions

- ทำไม network architecture model มีประโยชน์
- OSI มีกี่ชั้น อะไรบ้าง และ บรรยายหน้าที่ของแต่ละชั้นพอสังเขป

Computer and Data Communications

Chapter 1_3

Internet Model

โดย เศรษฐา ตั้งค้าวานิช

Objective

- List the layers of the Internet model and describe the duties of each layer
- Compare the OSI and Internet model and list difference and similarities

Internet Model

- Internet Model incorporate the TCP/IP protocols
- Internet Model has surpassed the OSI in popularity and Implement

Internet Model

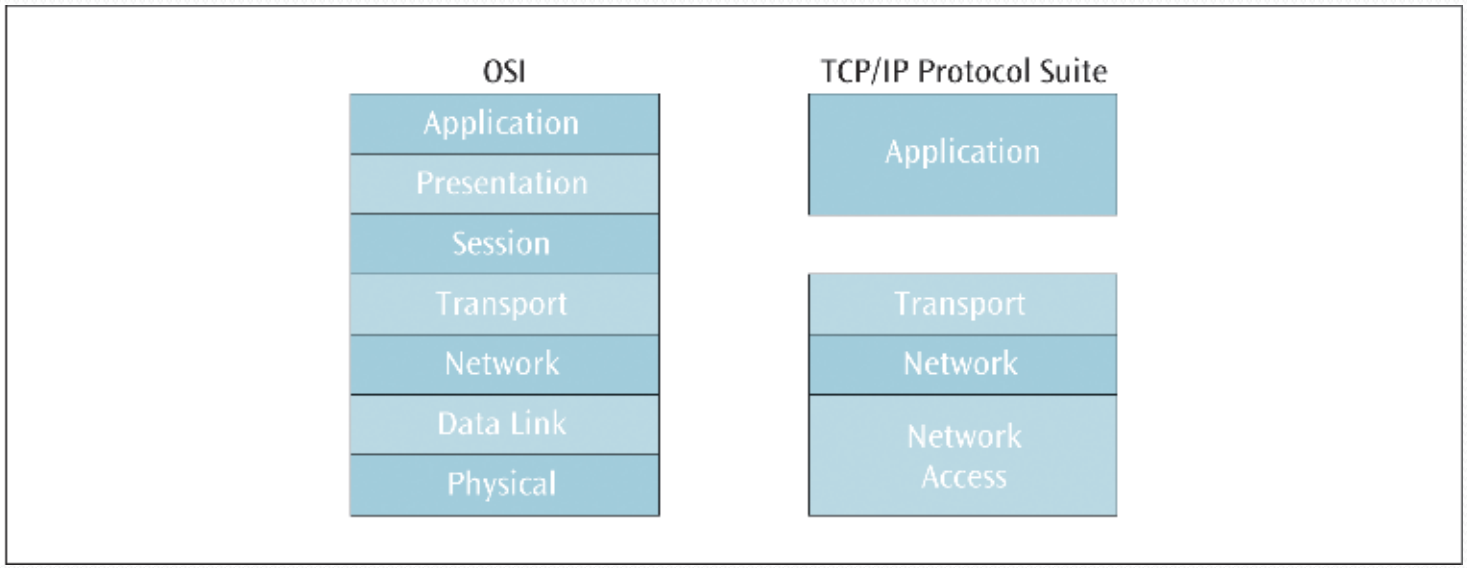


Figure 1-14
The layers of the TCP/IP protocol suite compared to the layers of the OSI model

Internet Model

- Application layer - equivalent to the OSI's presentation and application layers
- Example of Applications at Application layer
  - File Transfer Protocol(**FTP**)is used to transfer files from one computer system to another.
  - **Telnet** to allow a remote user to log into another computer system.
  - **HTTP** to allow browser and server to send and receive www pages.

Internet Model

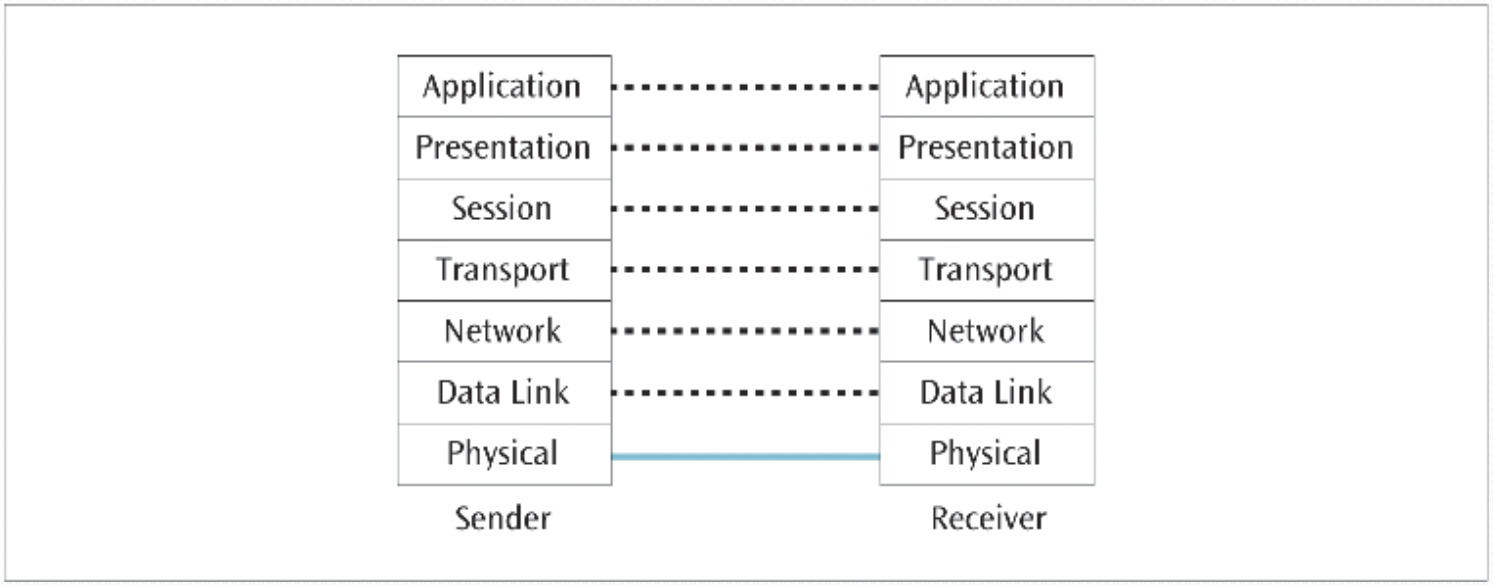
- Transport layer - performs same function as OSI transport layer
- Transport layer use Transport Control Protocol (**TCP**) and User Datagram Protocol(**UDP**)
- Network (Internet or internetwork) layer - roughly equivalent to the OSI's network layer
- One major distinction is that Internet use the Internet Protocol (**IP**)almost exclusively to transfer data between networks

Internet Model

- Network access (data link/physical) layer - equivalent to the OSI's physical and data link layers

Logical and physical connections

Figure 1-16
Sender and receiver communicating using the OSI model

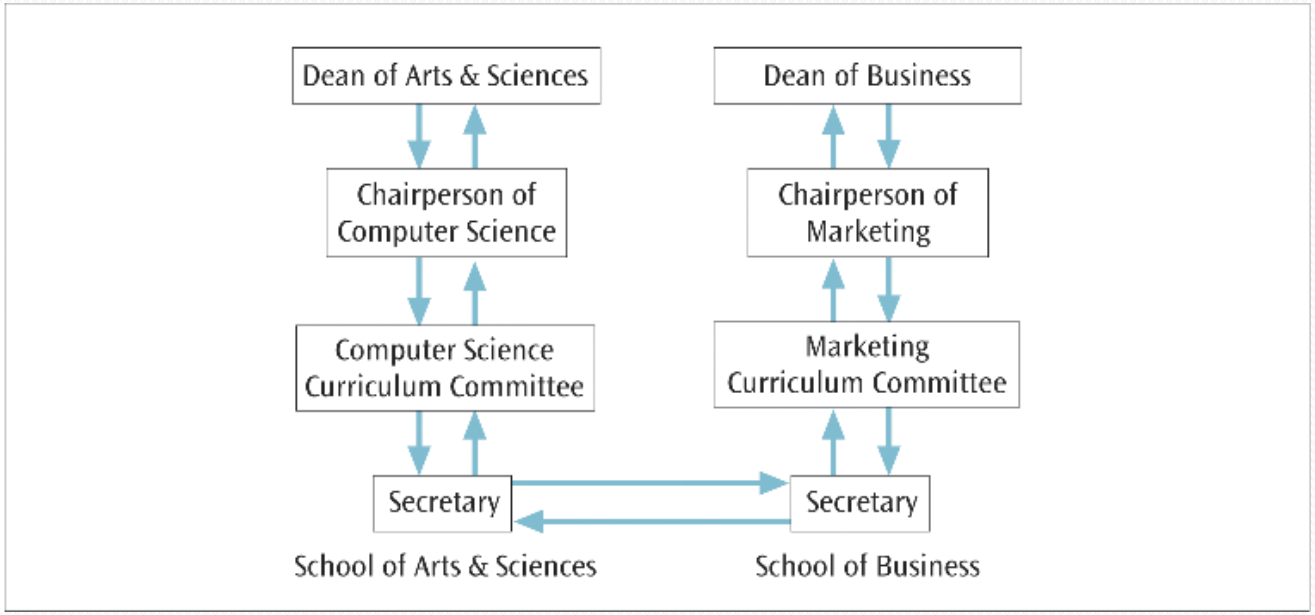


Logical and physical connections

- A logical connection is one that exists only in the software, while a physical connection is one that exists in the hardware.
- A logical connection is a flow of ideas, without a direct physical connection, between sender and receiver at a particular layer.
- Note that in a network architecture, only the lowest layer contains a physical connection, while all higher layers contain logical connections.

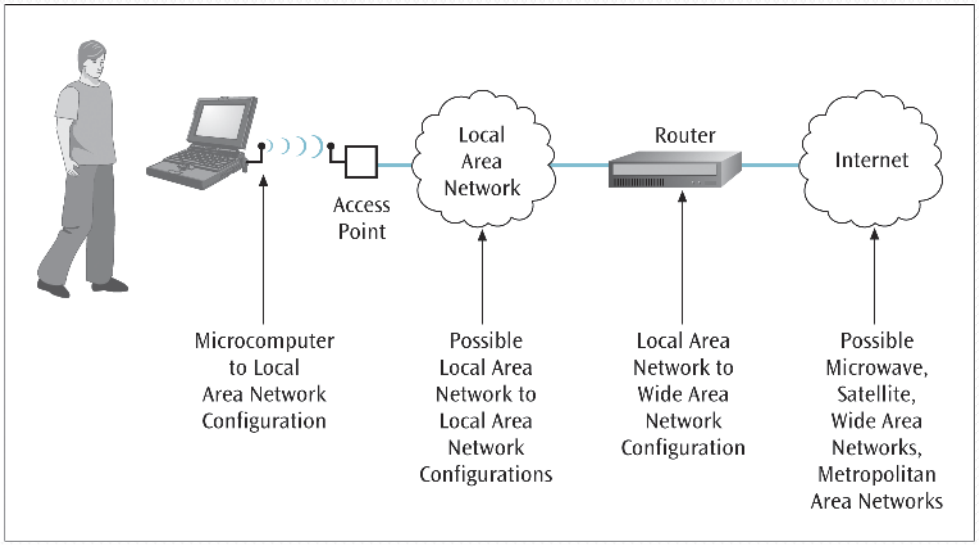
Example of data flow through layers

Figure 1-17
Flow of data through the layers of bureaucracy



Network Configurations in Action

Figure 1-18
The numerous network configurations involved with a user downloading a Web page at work



The TCP/IP Protocol Suite in Action

- Note the flow of data from user to web browser and back
- At each layer, information is either added or removed, depending on whether the data is leaving or arriving at a workstation
- The adding of information over pre-existing information is termed *encapsulation*

The TCP/IP Protocol Suite in Action

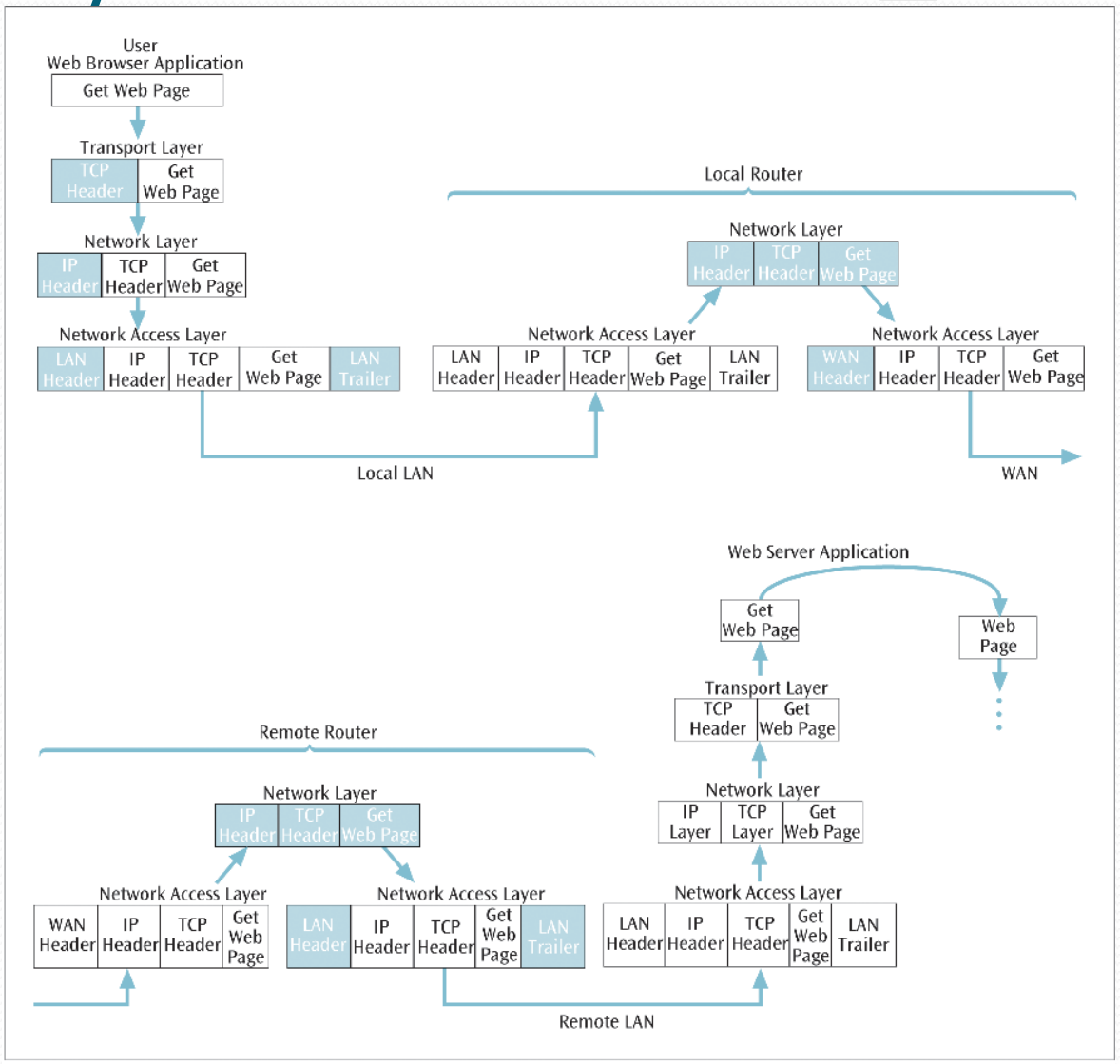


Figure 1-19
Path of a Web page request as it flows from browser to Internet server and back

Review Questions

- Internet model มีกี่ชั้น อะไรบ้าง และ บรรยายหน้าที่ของแต่ละชั้นพอสังเขป
- ชั้นต่างๆของ OSI model สามารถเปรียบเทียบกับชั้นของInternet model ได้อย่างไร
- Application อะไรบ้างที่ใช้อย่างแพร่หลายใน Internet model
- อะไรคือความแตกต่างระหว่าง logical และ physical connection